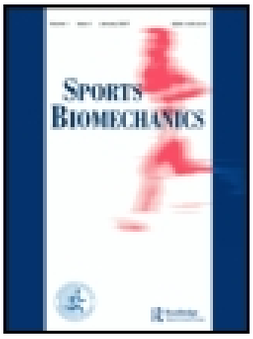

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Acceleration profile of an acrobatic act during training and shows using wearable technology

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ABSTRACT
The purpose of this study was to describe the mechanical characteristics of a trampoline circus act and its individual tracks performed in training and shows using a tri-axial accelerometer. A track is an artist's specific role within a choreographed act. Seven male acrobats performed their trampoline act during training and shows while wearing a triaxial accelerometer and reported ratings of perceived exertion (RPE) after each trial. Average acceleration (AVG), root mean square (RMS), root mean to the fourth (RM4), time spent in specific acceleration ranges and RPE were measured/recorded from training and show acts. Paired *t*-tests compared dependent variables between training and show. Acceleration AVG, RMS and RM4 were significantly higher ($p < 0.05$) in training than show. RPE was significantly higher ($p < 0.05$) in show than training. No significant differences existed in time spent in any of the acceleration ranges between training and show. GPS devices have been used to manage workloads in field sports but are inoperable in theatres. But, inertial measurements may be an effective alternative to describe mechanical demands in theatre or arena environments. Wearable technology may be useful to coaches to improve understanding of track demands to manage artist workloads.

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Acrobatics; performance; kinematics

Introduction

Athletes are employed for circus performances from a variety of disciplines closely related to sports such as gymnastics, diving, martial arts and acrosport (Shrier & Halle, 2011). Many circus performers are elite ex-athletes from a primary sport (e.g., gymnastics, diving, etc.). Similar to these sporting disciplines, workload management and awareness is critical to maintain high performance and reduce the risk of injury due to overtraining (Evans, Evans, Carvajal, & Perry, 1996; Grapton, Lion, Gauchard, Barrault, & Perrin, 2013; Kirialanis, Malliou, Beneka, & Giannakopoulos, 2003; Munro, 2014; Shrier & Halle, 2011; Shrier et al., 2009; Wanke, McCormack, Koch, Wanke, & Groneberg, 2012). Many circus artists have unique skillsets and contributions to their show. Each show is comprised of a series of acts. Each act is comprised of concurrent tracks that individual artists are responsible for

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performing. Artists may be assigned different tracks throughout the week based on artist availability and physical readiness. Therefore, losing artist availability is a major deterrent to the quality of the show. To fill an empty track, other artists may have to extend their role to support the needs of the show, causing sudden changes in workload.

Cirque du Soleil is a global leader in live entertainment renowned for its high-quality, artistic performances. The Quebec-based troupe currently has 18 shows performing worldwide with almost 1,300 artists. Cirque du Soleil artists typically perform 10 shows a week with only a few weeks off throughout the year. A better understanding of individual track demands could improve management of artist workloads. Unfortunately, there is limited research on the biomechanics (Grapton et al., 2013; Helten, Brock, Müller, & Seidel, 2011) and physiology (Jensen, Scott, Krustrup, & Mohr, 2013) of acrobatic performances during training or shows, which challenges our ability to categorise and/or quantify track demands or intensity. Furthermore, defining and/or measuring track intensity is complicated because of a large spectrum of athleticism, fitness and experience among acrobatic artists.

In recent years, accelerometers and inertial sensors have been used to quantify movement during other athletic competition and training sports, such as rugby, hockey and soccer (Boyd, Ball, & Aughey, 2013; Cummins, Orr, O'Connor, & West, 2013; Polglaze, Dawson, Hiscock, & Peeling, 2015). These devices allow for quantifying movements that are not suited to laboratory testing. Teams aim to optimise athlete workload by quantifying physical demands in their sport, but this has never been done in circus performers. Therefore, the purpose of this study was to use a tri-axial accelerometer to describe the mechanical characteristics of a trampoline circus act and its individual tracks performed in training and shows. This study provides the methods used to describe movements with the intent of quantifying individual track demands for workload management. We had two hypothesises, (1) that no differences in acceleration profiles would exist between training and shows, and (2) large variations would be observed among tracks. However, it is important to emphasise that the main objective of this work was descriptive in nature.

Methods

Experimental approach

This study was a non-experimental descriptive study. We measured tri-axial accelerations using a commercial product (Hexoskin, Carre Technologies Inc, Montreal, Québec, Canada) with a tri-axial accelerometer located on the lateral right hip. Seven acrobats performed 3 training acts on separate days, and 2 show acts on one day. All testing occurred within one week. To gather information about the subjective intensity of the act for each participant, participants reported their rating of perceived exertion (1–10 scale) following the completion of the act or show.

The act was comprised of 9 tracks coordinated to music. The stage was comprised of three rectangular trampolines (1.83 m × 3.66 m) in a row on the ground with the middle trampoline rotated 90°, and a fourth trampoline suspended in mid-air (5.18 m high) that reflects the orientation of the middle trampoline on the ground. Two scalable walls (5.49 m high) on either side of the ground-level trampolines create the perimeter for the act.

The act is choreographed to a song approximately 5 min long. Each artist starts in different positions on stage: five artists begin on the ground, three artists begin on the suspended

trampoline and one artist begins on the top of the wall. For the first 40 seconds, minimal movement occurs (walking, posturing, stillness) until the first trampoline contact by two artists falling from the suspended trampoline. Then, throughout the remainder of the act, there are 1–5 artists interacting with the trampolines while the other artists perform choreographed walking, dancing or waiting for their cue.

Participants

Seven male acrobats (Height: 1.75 ± 0.05 m; Mass: 78.8 ± 4.5 kg; Age: 28.9 ± 3.3 years; Body Composition: 15.5 ± 3.9% body fat) participated in all testing sessions (3 in training, 2 in shows). Each participant provided informed consent prior to participation in the study, which was approved by the University of Nevada Las Vegas' Institutional Review Board (#724768-4).

Instrumentation

The Hexoskin is a tight-fitting sleeveless shirt that collects heart rate and breathing data from sensors located in the chest and abdominal region in addition to tri-axial accelerations from an accelerometer in a pocket at the lateral right hip. Only accelerometer data were used for this study. The accelerometer unit also stores data which are uploaded to a proprietary user interface via Hexoskin's website where data can be viewed and downloaded for analysis. A trial of data was initiated when the shirt was physically connected to the accelerometer unit via a connecting cable. Once unplugged, the trial was stopped with data stored within the unit.

The accelerometer device was 13 × 42 × 72 mm in size and weights 40 g. It was secured in a pocket on the right hip of the compression shirt, which provides a snug fit (Figure 1). The acceleration range is ±16 g with data collected at 64 Hz. Hexoskin's proprietary software down samples the data to 1 Hz by averaging each second of data. The data were downloaded from Hexoskin's user interface for analysis using custom laboratory software.

Figure 1. Hexoskin shirt with accelerometer unit.
Notes: Connecting the cable to the accelerometer unit initiated data collection. Unplugging the cable ended the data collection. The accelerometer unit is placed within a pocket sewn into the shirt.

Procedures

Artists performed the act during three training sessions and two shows. Each artist performed one of the nine tracks for all sessions and shows. Prior to training sessions, participants would warm up with a variety of trampoline jumps for approximately 20 min. Once their warm up was complete, all participants plugged in and out of the Hexoskin unit at the same time. Data from two of the participants were not collected because their acrobatic movements led to accidentally unplugging the unit. Unfortunately, recollecting trial attempts were limited due to show and artist schedule constraints.

Data reduction

To focus the analysis on individual tracks, data were extracted by visually inspecting the acceleration profiles to find distinct start and end points of movement. The process of selecting the start/end points was based on inspection of magnitude of acceleration profiles and identifying a start acceleration of approximately 0.25 g given that the length of the track was approximately known as well as the type of movements in the beginning and ending point of each track. Identifying start/end points was performed by the same researcher for all participants. This approach resulted in similar length of data reflecting the time of the entire act, but each data-set was a unique length. We performed a paired t-test on length of data-sets and there was no difference between training and show ($p > 0.05$). Therefore, analysis was done using actual data-set length; however, data-sets were time normalised for presentation purposes.

A custom MATLAB program (The MathWorks, Natick, MA, USA) was used to analyse the Hexoskin-processed data. Resultant Acceleration was calculated ($\sqrt{(x^2 + y^2 + z^2)}$) from the three axes of the accelerometer. Therefore, acceleration values were always positive. Resultant acceleration was then processed to calculate average magnitude (AVG), root mean square acceleration (RMS), root mean 4th root (RMS4), time spent in acceleration ranges and standard deviations of acceleration within training sessions or shows.

All variables were calculated during training sessions (AVGtrain, RMStrain, RM4train) and shows (AVGshow, RMSshow, RM4show), and averaged for training and shows for each participant. For time spent in acceleration range, acceleration magnitude was classified into five ranges: $0 \leq$ Very Low (VL) < 0.1 g; $0.1 \leq$ Low (L) < 0.3 g; $0.3 \leq$ Moderate (M) < 0.6 g; $0.6 \leq$ High (H) < 1; $1 \leq$ Very High (VH) ≤ 16 g. Time spent in each range was averaged for training (tVL, tL, tM, tH, tVH) and shows (sVL, sL, sM, sH, sVH), in addition to total time for training and shows (tTIME, sTIME). Reported ratings of perceived exertion (0–10 scale) were averaged for training (tRPE) and shows (sRPE).

Paired-sample *t*-tests were used to compare each variable between training and show. Effect sizes were calculated for the acceleration ranges between training and show and interpreted according to Cohen: trivial (0–0.2), small (0.2–0.5), moderate (0.5–0.8), large (>0.8) (Cohen, 1988).

Results

AVGtrain was significantly higher than AVGshow (0.53 ± 0.12 g vs. 0.47 ± 0.10 g; $p = 0.030$), Figure 2). RMStrain was significantly higher than RMSshow (0.85 ± 0.15 g vs. 0.77 ± 0.14 g;

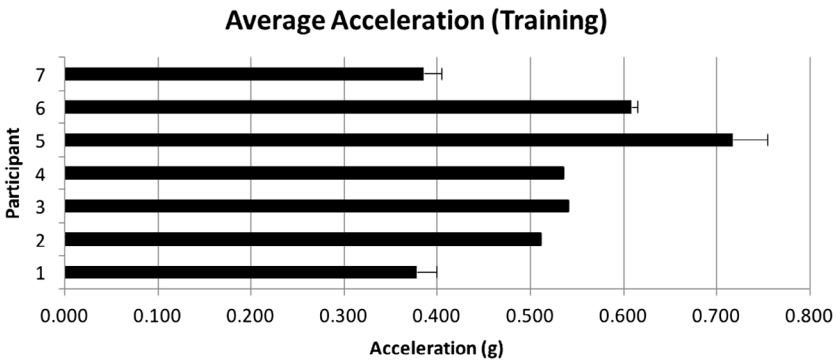

Figure 2. Average accelerations in training (standard error) for each participant.

Table 1. Time spent (s) in very low (0–0.1), low (0.1–0.3), medium (0.3–0.6), high (0.6–1.0), and very high (>1) acceleration ranges.

Acceleration range	Time in training (s)	Time in show (s)	*p*-value	Effect size
Very low	85.8 ± 35.1	93.1 ± 37.3	0.210	0.20
Low	82.7 ± 19.4	85.1 ± 20.3	0.690	0.12
Moderate	68.9 ± 23.7	67.4 ± 26.2	0.700	0.05
High	35.3 ± 17.6	27.9 ± 9.3	0.150	0.55
Very high	47.9 ± 12.9	41.0 ± 10.3	0.050	0.55

Note:
Cohen's effect size interpretations: trivial (0–0.2), small (0.2–0.5), moderate (0.5–0.8), large (>0.8).

$p = 0.010$). RM4train was significantly higher than RM4show (1.40 ± 0.22 g vs. 1.23 ± 0.21 g; $p < 0.010$). tRPE was significantly lower than sRPE (2.76 ± 0.53 vs. 3.43 ± 1.10; $p < 0.030$).

There was no significant difference in the time spent in any acceleration range (VL, L, M, H, VH) between training and shows ($p > 0.05$, Table 1, Figure 3(a) & (b)).

The tTIME and sTIME were not significantly different ($p = 0.080$), indicating similar lengths of data-sets during training and show. Exemplar data of the highest (participant 5) average accelerations is displayed in Figure 4(a) & (b). A comparison of time-normalised acceleration during acts from one training session and one show is depicted in Figure 5.

Discussion and implications

Main observations

The purpose of this study was to describe the mechanical characteristics of a trampoline act and the individual tracks in training and shows using a triaxial accelerometer. The main observation of this descriptive study was higher average accelerations in training compared to shows with a moderate effect size 0.53. The acceleration data in this study are similar to work done by Helten et al., 2011 who classified trampoline jumps based upon data from inertial sensors. The authors reported peak accelerations of about 4–5 g, whereas peak accelerations in our study were about 3.5 g. Helten et al. (2011) used multiple sensor locations and collected accelerometer data at higher sampling frequencies (100 Hz) than what was used in the present study (single 3D accelerometer with a sample rate: 1 Hz). In the present study, accelerations tended to peak during contact with the trampoline.

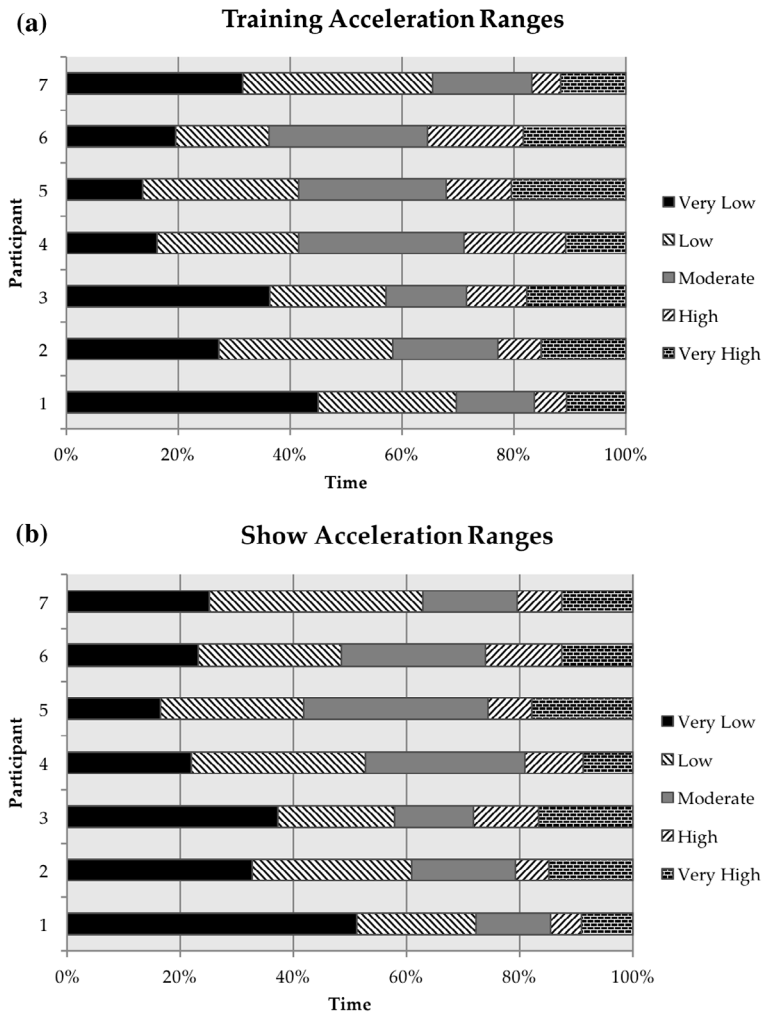

Figure 3. (a) Acceleration ranges during training as a percent of act length. Ranges are depicted from left (Very Low) to right (Very High), (b) Acceleration ranges during show as a percent of act length. Ranges are depicted from left (Very Low) to right (Very High).

Helten et al. used a threshold of 35 m/s^2 (3.56 g) to distinguish the trampoline contact phase, which could vary ±5 m/s^2 (0.51 g) for different jumps, trampolines or skill levels. Therefore, using accelerometers during acts could be useful in calculating the number of trampoline contacts, which may allow a calculation of plyometric volume, an important measure to understand for effective workload management. However, many factors during an acrobatic trampoline manoeuvre influence acceleration measurement: angular velocity of the person, size and stiffness of the trampoline and height of the jump (Helten et al., 2011). Therefore, volume calculations or estimations may be limited (i.e. a single acceleration threshold may not catch all jumps) in a circus environment due to the variety of rotational and height demands.

When the acceleration data were divided into acceleration ranges, no significant difference in time spent in a range was observed between training and show for any of the

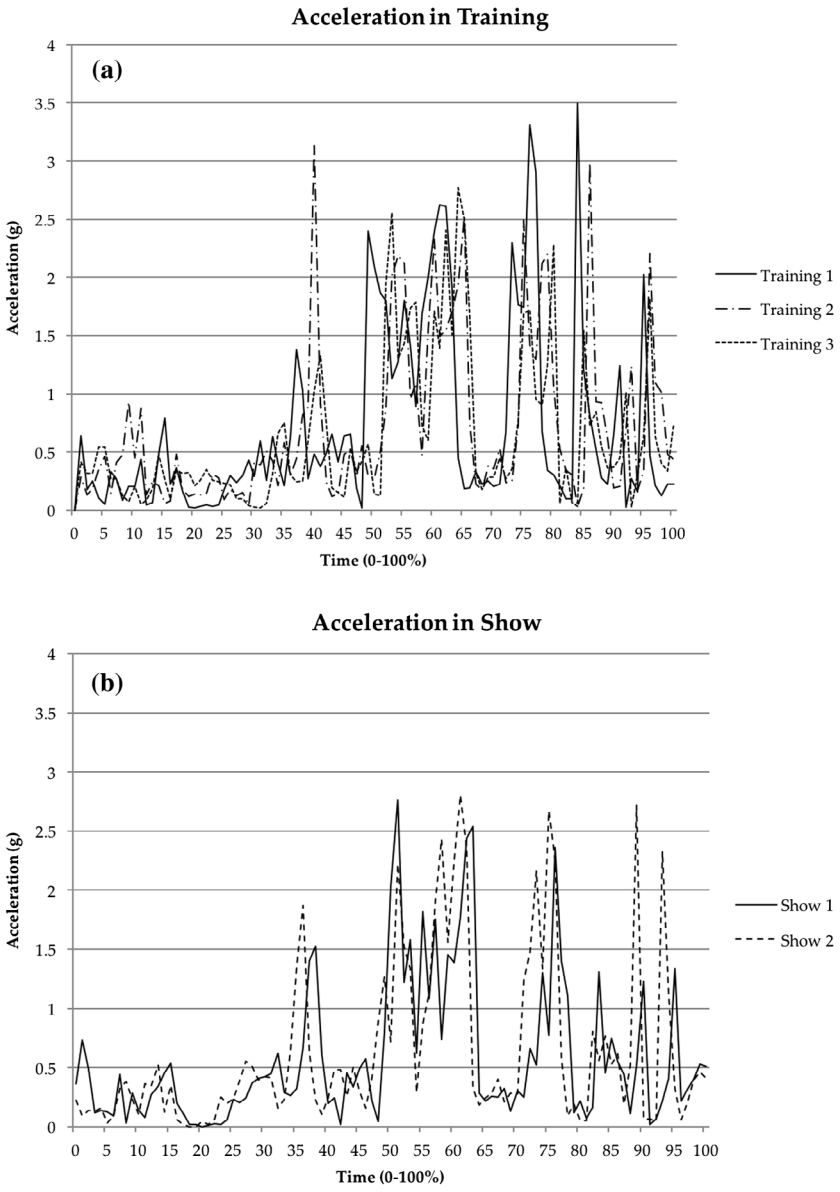

Figure 4. (a) Participant 5 time-normalised acceleration in training (3 trials), (b) Participant 5 time-normalised acceleration in shows (2 trials).

acceleration ranges. However, effect sizes of the H and VH acceleration ranges (Table 1) suggest some of the time spent in the H and VH acceleration range during training decreased during shows, which may indicate technique is being prioritised over movement intensity for short durations of the act.

Although this study was descriptive in nature, some conjectures about movements can be made. Perhaps artists are more focused during the show, filter out unnecessary movements and therefore accelerations are less. Or, the environment of the show causes fatigue and

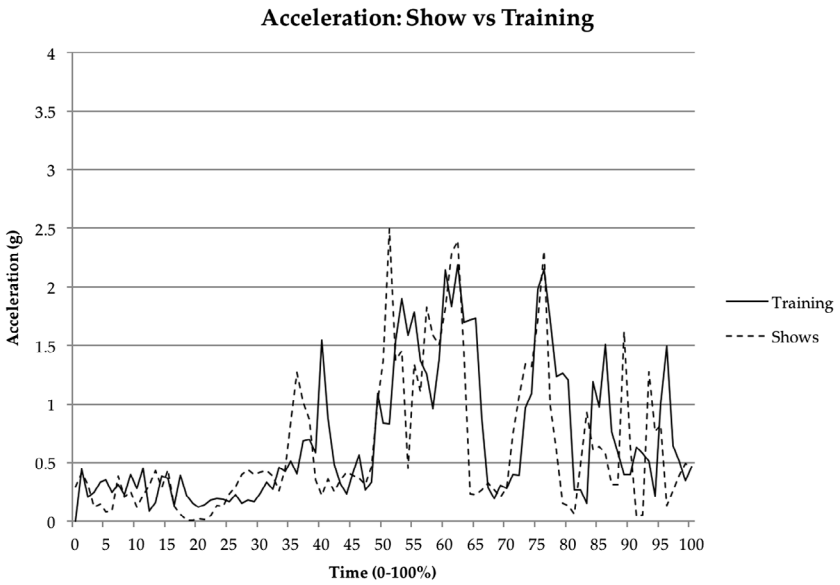

Figure 5. Participant 5 Training vs. Show time-normalised acceleration.

influences pacing strategies to safely perform their task in sync with co-artists. In support of this notion, the RPE values were significantly higher in training than shows. The training environment may be less constrained, permitting confidence to attempt higher intensity manoeuvres at the expense of synchronisation.

Circus is one of the only industries where artists or athletes perform shows more frequently than they are training. Furthermore, the best performers must be on stage to deliver the best show. The results of this study demonstrate clear differences in average accelerations across tracks, which could support coaching decisions related to artist track management. For example, an artist performing an intense track for a period of months could be cycled through less demanding tracks to support recovery, whereas an artist returning from injury could perform easier tracks and progress to harder tracks over time. Alternatively, understanding the most difficult tracks would help coaches place their artists into manageable tracks and avoid asking an artist to overextend their abilities.

The current literature investigating workload management has not included circus disciplines, but rather field sports because of the utility of Global Positioning Systems (GPS) (Cummins et al., 2013). However, GPS is limited regarding change of direction, so accelerometers have been used to describe workload intensity in greater detail (Coutinho et al., 2015; Ehrmann, Duncan, Sindhusake, Franzsen, & Greene, 2016). The evidence in this area of study has supported the notion that sudden increases in acute workload, as determined by GPS alone (Hulin, Gabbett, Caputi, Lawson, & Sampson, 2016) and combined with accelerometers (Ehrmann et al., 2016), increases the risk of injury. For example, average acceleration multiplied by body mass and normalised to time could be used to describe the intensity of movement. Furthermore, high chronic workloads are reported to produce a protective effect against sudden increases in acute workloads (Hulin et al., 2016), which suggests athletes should train harder, *progressively*, to gain this protective effect (Gabbett, 2016). For example, very high acute/chronic workload ratios may exceed 1.5 while an

optimal acute/chronic workload may be around 0.8–1.3 (Gabbett, 2016), which could be used to guide progressive overload throughout a periodised training program. If coaches consider these concepts within the circus framework of choreography, each track or act within a show could be assigned volume and intensity demands to support coaching decisions about artist rotations, return from injury and new artists.

A limitation of this study is that one act was analysed. It is important to recognise the variety of circus disciplines before generalising the results of this study. Acts in various shows may contain more homogenous or heterogeneous disciplines than the trampoline act in this study. Furthermore, fitness levels of artists must be considered. An artist who regularly performs jumping movements may have difficulty performing hanging movements, for example, so the mechanical demands may need to be supported by internal workload measures like heart rate and/or subjective ratings of perceived exertion. Some acts may contain improvisation phases, audience interaction, lack of music or light and completely different terrain. Many variables may affect circus performances, which requires the coach to attempt these methods for themselves to figure utility. It is important to recognise that mechanical demands, as measured by a single tri-axial accelerometer, cannot fully describe the workload of an artist. Furthermore, the sample rate of used may limit the ability to detect higher acceleration peaks than what we observed. It may be that important characteristics to describe movements are more related to peak accelerations. In that case, accelerometers with higher sample rates would be important to use. However, it is also important to explore the ability of using available wearable technology to measure movements during an act.

Practical applications of using wearable technology

The dynamic environment of the show, and our results, could have differed if the instrument reported data at higher frequencies or provided time-synchronised data among devices. Many wearable technology devices are suited for group use. However, Hexoskin is designed for individual use rather than group use. Other data acquisition systems may address data synchronisation and sampling frequency challenges to allow more precise comparisons of acts and high intensity manoeuvres. Regardless, circus acts should be measured and interpreted individually since they are unique to their respective show. If coaches seek to analyse specific movements or sequences of movements, a higher sampling frequency and multiple data collection devices may be required to provide enough information to be useful. In addition, if coaches want to gain insight into their individual track comparisons, more trials across multiple tracks may need to be performed by each artist to rank track intensities appropriately.

The methods used in this study could be modified for a variety of acrobatic sports, such as competitive trampoline, tumbling, gymnastics and dance. Since these activities are typically choreographed, accelerometers could further a coach's or artist's understanding of act demands and artistic athleticism. Acceleration profiles of acts may be valuable to support workload management and performance assessment in the performing arts.

Limitations

Acrobatic movements occur at frequencies greater than 1 Hz. Therefore, in accordance with Nyquist's Sampling Theorem, acceleration values such as rates and peaks were not

considered for description of individual movements. However, the inability to describe accelerations of single movements in detail is accompanied by affordability, portability and practicality for prolonged data collection to describe general movement outside of a laboratory setting. For these reasons, we found the Hexoskin suitable for this study. However, transitional time remains between the plug in/out and the actual start/end points of the act in both training and shows. In addition, each track has different start and end points within the act. For example, some tracks require little to no activity at the beginning of the act while others begin activity immediately. We do not believe this has a large effect on the descriptive outcomes and conclusions of the study.

Conclusion

In conclusion, our hypothesis was confirmed that average acceleration did not differ between shows and training. According to effect sizes, we found this trampoline act to have moderately different average accelerations in training and shows, but the time spent in specific ranges of accelerations was not different at any level. This information may help coaches manage their artist workloads by assigning intensities based on acceleration data to various tracks and acts. In practical application, coaches could use commercial accelerometers to gain insight on the mechanical demands of their acts in the circus environment, which may be particularly useful in determining track intensities to guide artist rotations. Considering the rigorous year-round schedule of circus performers, improving any aspect of workload management could lead to longer careers and fewer missed shows. Wearable technology has become a global market among sporting organisations for these purposes. However, it remains to be determined with long-term studies if a coach can improve performance over time if they implemented, interpreted and made data-driven decisions.

Ethical approval statement

This study was approved by the University of Nevada Las Vegas Institutional Review Board, #724768-4.

Acknowledgements

We would like to thank the coaches and artists at Cirque Du Soleil for their efforts in making this study possible.

Disclosure statement

Leland Barker and Dr. John Mercer report an academic partnership with Cirque Du Soleil. This manuscript was prepared as part of an academic partnership between the University of Nevada Las Vegas and Cirque Du Soleil. The views expressed in this manuscript are our own and not an official position of the institutions involved.

References

Boyd, L. J., Ball, K., & Aughey, R. J. (2013). Quantifying external load in Australian football matches and training using accelerometers. *International Journal of Sports Physiology and Performance, 8*, 44–51. doi:10.1123/ijspp.8.1.44.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Hilsdale, NJ: Lawrence Erlbaum.

Coutinho, D., Gonalves, B., Figueira, B., Abade, E., Marcelino, R., & Sampaio, J. (2015). Typical weekly workload of under 15, under 17, and under 19 elite Portuguese football players. *Journal of Sports Sciences, 33*, 1229–1237. doi:10.1080/02640414.2015.1022575.

Cummins, C., Orr, R., O'Connor, H., & West, C. (2013). Global positioning systems (GPS) and microtechnology sensors in team sports: A systematic review. *Sports Medicine, 43*, 1025–1042. doi:10.1007/s40279-013-0069-2.

Ehrmann, F. E., Duncan, C. S., Sindhusake, D., Franzsen, W. N., & Greene, D. A. (2016). GPS and injury prevention in professional soccer. *The Journal of Strength & Conditioning Research, 30*, 360–367. doi:10.1519/JSC.0000000000001093.

Evans, R. W., Evans, R. I., Carvajal, S., & Perry, S. (1996). A survey of injuries among broadway performers. *American Journal of Public Health, 86*, 77–80. doi:10.2105/AJPH.86.1.77.

Gabbett, T. J. (2016). The training-injury prevention paradox: Should athletes be training smarter and harder? *British Journal of Sports Medicine, 50*, 273–280. doi:10.1136/bjsports-2015-095788.

Grapton, X., Lion, A., Gauchard, G. C., Barrault, D., & Perrin, P. P. (2013). Specific injuries induced by the practice of trampoline, tumbling and acrobatic gymnastics. *Knee Surgery, Sports Traumatology, Arthroscopy, 21*, 494–499. doi:10.1007/s00167-012-1982-x.

Helten, T., Brock, H., Müller, M., & Seidel, H. (2011). Classification of trampoline jumps using inertial sensors. *Sports Engineering, 14*, 155–164. doi:10.1007/s12283-011-0081-4.

Hulin, B. T., Gabbett, T. J., Caputi, P., Lawson, D. W., & Sampson, J. A. (2016). Low chronic workload and the acute: Chronic workload ratio are more predictive of injury than between-match recovery time: A two-season prospective cohort study in elite rugby league players. *British Journal of Sports Medicine, 50*. doi:10.1136/bjsports-2015-095364.

Jensen, P., Scott, S., Krustrup, P., & Mohr, M. (2013). Physiological responses and performance in a simulated trampoline gymnastics competition in elite male gymnasts. *Journal of Sports Sciences, 31*, 1761–1769. doi:10.1080/02640414.2013.803591.

Kirialanis, P., Malliou, P., Beneka, A., & Giannakopoulos, K. (2003). Occurrence of acute lower limb injuries in artistic gymnasts in relation to event and exercise phase. *British Journal of Sports Medicine, 37*, 137–139. doi:10.1136/bjsm.37.2.137.

Munro, D. (2014). Injury patterns and rates amongst students at the national institute of circus arts: An observational study. *Medical Problems of Performing Artists, 29*, 235–240.

Polglaze, T., Dawson, B., Hiscock, D. J., & Peeling, P. (2015). A comparative analysis of accelerometer and time-motion data in elite men's hockey training and competition. *International Journal of Sports Physiology & Performance, 10*, 446–451.

Shrier, I., & Halle, M. (2011). Psychological predictors of injuries in circus artists: An exploratory study. *British Journal of Sports Medicine, 45*, 433–436. doi:10.1136/bjsm.2009.067751.

Shrier, I., Meeuwisse, W. H., Matheson, G. O., Wingfield, K., Steele, R. J., Prince, F., & Montanaro, M. (2009). Injury patterns and injury rates in the circus arts: An analysis of 5 years of data from cirque du soleil. *The American Journal of Sports Medicine, 37*, 1143–1149. doi:10.1177/0363546508331138.

Wanke, E., McCormack, M., Koch, F., Wanke, A., & Groneberg, D. A. (2012). Acute injuries in student circus artists with regard to gender specific differences. *Asian Journal of Sports Medicine, 3*, 153–160.